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MILITARY SPECIFICATION

WINCH, DIESEL-DRIVEN, 2-DRUM, 5.9 TON PULL, WITH BOOM SWINGER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a winch (double drum) that exerts a 5.9 ton pull powered by a diesel engine, for general use and with a 30-ton, stiff-leg derrick.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, Flat and Seals.
RR-W-410	- Wire Rope and Strand.
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated Adhesive, Pressure Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1400	- Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS 24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS52131	- Connector, Plug, Electrical, Inter-Vehicle Power Cable.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

ANSI/AWS D1.1 - Structural Welding Code, Steel, Sections 5 and 6, Welding Qualifications.
D14.3 - Earthmoving and Construction Equipment Welding, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Standard 114 - Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The winch shall consist of a diesel engine, double drum, manual transmission, pneumatic controls, and boom swinger. In addition, the winch shall have all required shafting, drum brakes, drum friction clutches, reduction gearing, ratchets, pawls, bearings, guards, and all other parts necessary for its performance and intended use except for cables when not required (see 3.11.6). The winch shall be mounted on a skid base to assure its alinement is maintained under all conditions of operation.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The winch shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-C-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the winch may be newly fabricated from recovered materials to the maximum extent practicable, provided the winch produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the winch.

3.3.3 Bearings. Unless otherwise specified herein, rotating parts shall be mounted on ball, roller, or tapered bearings to the extent normally provided in standard commercial practice.

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3.3.4 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fasteners such as bolts, nuts, and screws shall be held to a minimum number of types and sizes.

3.3.5 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.4 Environmental conditions.

3.4.1 Operational temperatures. The winch shall not be damaged during operation and transport in any ambient air temperature between -25° F and +110° F.

3.4.2 Precipitation. The winch shall not be damaged or be rendered inoperative when exposed to precipitation for not less than 15 minutes ranging from a drizzle to a heavy rain of 1/4 gallon per minute per square foot of wetted surface impinging on the winch with a 30 mph wind. During and after the precipitation, the winch shall operate as specified herein.

3.4.3 Storage. When in outside storage, the winch shall withstand climatic extremes from -28° F to +145° F, without deterioration that may cause failure of the winch or of any component.

3.5 Safety. All rotating and reciprocating parts and parts subject to high temperature shall be guarded when such parts are exposed to contact by operator and maintenance personnel performing daily maintenance functions. Nonfunctional edges shall be rounded, projecting points shall be blunted or rounded, and excessive length of fasteners shall be avoided. Steps and platforms shall have antiskid, ice resistant surfaces. Oil level indicators for engine, transmission, and other components shall be within easy and safe access of the operator or maintenance personnel.

3.6 Human factors. The characteristics of the winch shall provide for operation and maintenance by personnel ranging from the small person clothed, through the large person arctic clothed, in accordance with SAE J833, SAE J925, and MIL-STD-1472 as specified herein.

3.6.1 Noise limits. The noise produced by the winch shall not exceed 85dB(A) at the operator's station when tested in accordance with 4.5.2.11. If technical infeasibility is established in accordance with MIL-STD-1474, or the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85dB(A) limit. The precautions of 6.4 shall be provided if, and only if, MIL-STD-1474 procedures have been pursued and documented to the satisfaction of the procuring activity. Hazard signs, if required, shall conform to MIL-STD-1474 and be readable at the distance indicated on the sign.

3.7 Maintainability. Provision shall be made for adjustment, servicing or replacement of parts and components. When access openings are used on the exterior of the winch, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover whenever it is required. Dimensions

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of hand or arm access openings shall be in accordance with MIL-STD-1472 for arctic clothing. Filters for the fuel and engine oil shall be accessible for servicing. Each operation shown on the following maintenance operations list shall be accomplished by one person using common tools and special tools furnished with the winch within the time limit specified. Upon completion of the maintenance operations the winch shall operate as specified herein.

Maintenance operation list:	Time allowed:
a. Remove, replace, and adjust main drive and accessory drive belts or chains (including guards).	30 minutes for each set
b. Remove and replace alternator and readjust belt tension.	30 minutes
c. Disconnect battery cables, remove and replace battery, and reconnect cables.	20 minutes
d. Drain, change filters, and refill engine oil lubricating system.	45 minutes
e. Remove and replace fuel filters elements.	30 minutes.
f. Grease and lubricate the winch in accordance with the winch lubrication chart.	30 minutes
g. Remove and replace engine.	1 hour
h. Align engine, countershaft and drum.	45 minutes
i. Remove and replace brake lining.	1 hour
j. Remove and replace clutch lining.	1 hour

3.8 Performance. The winch shall conform to the performance requirements specified herein without malfunction, damage or permanent deformation when:

a. The winch skid is anchored to an immovable object.
b. The winch is equipped with all attachments, accessories, and equipment and is operated as specified herein.
c. The winch is completely serviced and adjusted.

3.8.1 Transport. The winch shall be transportable by skidding the winch over the terrain by manual means and by vehicle towing. When tied down, the winch shall be capable of transport by vehicle. The winch shall be capable of winching itself into and out of truck bodies, when suitable truck body skid ramps are used.

3.9 Diesel engine. For the purpose of selecting the engine, the winch shall be classified as a class I end item in accordance with MIL-STD-1410. The engine, including all systems, components and accessories, shall conform to MIL-STD-1410, class I except as otherwise specified herein. The engine shall conform to all operational requirements specified in table I of MIL-STD-1410. The power and speed rating of the engine shall be such that operations of the winch under any of the operational requirements specified herein will not require horsepower in excess of the continuous brake-horsepower rating of the engine as published by the engine manufacturer.

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3.9.1 Alternator. An alternator shall be provided and shall provide charge at engine low idle speed and shall maintain battery charge at full electric load. A regulator shall be furnished. The regulator may be integral with the alternator.

3.9.2 Charging receptacle. A charging receptacle conforming to MS52131 shall be furnished.

3.9.3 Engine mounting and housing. The engine shall be rigidly mounted on a sub-base which shall be mounted on the winch skid base. The engine shall be furnished with a housing.

3.9.4 Fuel tank. The fuel tank shall be of sufficient capacity for 8 hours of operation at the continuous brake horsepower consumption rate. A mechanically actuated fuel gage shall be furnished.

3.9.5 Cranking system. A 24-volt electrical cranking system shall be furnished.

3.9.6 Control systems. The following engine control systems shall be furnished:

a. High coolant temperature control of the warning light type.
b. Low lubricating oil pressure of the warning light type.
c. Overspeed control of the engine shut-off type. The warning light for controls (a) and (b) shall be mounted on the engine instrument panel.

3.9.7 Instrumentation. The engine shall be furnished with an instrument panel and shall be fitted with the following instruments:

a. Lubricating oil pressure gage.
b. Indicator, cooling liquid temperature.
c. Indicator, batery-generator.
d. Mechanical hourmeter.
e. Fuel pressure gage.

3.9.8 Engine cold starting aid. Unless otherwise specified (see 6.2), an engine cold starting aid (ether fuel primer) or a glow plug for each cylinder shall be furnished.

3.9.9 Batteries. Two batteries conforming to MS35000, type 6TN shall be furnished. Batteries shall be negative ground in accordance with SAE J538 and shall be readily accessible for servicing.

3.9.10 Battery box. A battery box shall be furnished. The interior of the box shall be protected with a coating conforming to TT-C-494, type II. The box shall have provisions for drainage and for gas venting.

3.9.11 Oil consumption. The average oil consumption of the engine during first article testing shall not exceed the limits specified in MIL-STD-1400, test method 1500, table II.

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3.10 Power transmission. Power shall be transmitted from the engine to a jackshaft on the winch by means of the clutch and a chain drive. The clutch shall be flywheel type and shall be mounted in a housing attached to the engine. The drive chain shall be enclosed and shall run in oil.

3.11 Winch and boom-swinger assembly. The winch assembly shall consist of a double-drum winch, mounted one drum in front of the other and individually controlled. The rear drum shall be elevated sufficiently above the front drum to provide clearance of the load line. Each drum shaft shall be equipped with a winch head. The boom-swinger assembly shall be constructed and installed in accordance with commercial standards for this type of equipment.

3.11.1 Winch. The winch shall exert a single-cable pull of not less than 11,800 pounds at a line speed of 200 feet per minute acting on the second wrap of the cable around the winch drum. The winch shall have a factor of safety not less than 4 based on the ultimate strength of the materials used in fabrication. When the winch is tested as specified in 4.5.2.6, there shall be no evidence of brakes fading.

3.11.1.1 Drums. Each drum shall be cast in one piece with integral brake flange, safety ratchet, and clutch flange. The drum shall be accurately machined and balanced for smooth operation. The drum barrel shall not be less than 14 inches in diameter by 24 inches long between flanges. The flange diameter shall be not less than 30 inches.

3.11.1.2 Clutch. Each drum shall be equipped with a clutch keyed or splined to the drum shaft and actuated by a hand control lever with connecting linkage. The clutch operating mechanism shall assure quick, easy action at all drum speeds without grab, chatter, or overheating and for quick, positive release without drag. The clutch lining shall be of sufficient size and capacity to transmit 125 percent of the rated engine output at full load without slippage.

3.11.1.3 Pawl and ratchet. Each drum shall be equipped with a pawl and ratchet that will hold the drum in any position when under full load with the brake released. Engagement and release of the pawl shall be actuated by a hand lever with connecting linkage that will automatically engage the pawl when pressure is released. Rotation of the drum in the takeup direction shall automatically release the pawl.

3.11.1.4 Winch head. The winch head shall be cast in one piece, accurately machined and balanced, and keyed to the end of the drum shaft on the opposite side of the hoist from the operator. The winch head shall be hardened to allow the use of wire rope and shall be suitable for use with a 4-inch nominal circumference manila rope.

3.11.1.5 Winch head line guide. Unless otherwise specified (see 6.2), a winch head line guide comprised essentially of a steel roller shall be mounted on a vertical shaft which is seated in a base plate and bolted to the winch (see figure 1). The inner side of the roller shall have a top and bottom sleeve containing anti-friction roller bearings permanently fastened to the roller to

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permit its rotation around the vertical shaft. The guide shall be positioned so that when the winch head is completely covered with rope, the inside edge of the guide shall line up with the outside end of the winch head. The line guide shall withstand a force of 1500 pounds applied at point of contact of the rope without damage or permanent deformation. A warning against tying off the line to the line guide shall be provided (see 3.19.1). The steel roller shall be a minimum of 4-1/2 inches in diameter. The length and vertical location shall be such that the top of the guide is at least 8 inches above the point of contact with the rope. The rope shall clear the base of the roller.

3.11.2 Boom swinger. The boom swinger shall consist of two boom-line drums mounted on a single shaft driven through a transmission system from the forward drum shaft. The swinger shall transmit a line pull of not less than 4,000 pounds at a rope speed of 50 feet per minute with the engine operating at maximum governed speed. When the boom swinger is tested as specified in 4.5.2.7, there shall be no evidence of brake fading.

3.11.2.1 Drums. Each drum shall be cast in one piece, accurately machined and balanced, and rigidly keyed to the boom-swinger shaft. The drum barrels shall be not less than 12 inches in diameter by 8 inches long between flanges.

3.11.2.2 Clutch and transmission. The boom swinger drive shall consist of disk or cone type clutches arranged to provide for forward and reverse rotation; a geared speed reducing transmission running in oil; and a silent chain drive running in oil. The clutches shall be accessible for adjustment.

3.11.3 Gears. All gears and pinions shall be of one-piece steel construction with cut teeth and shall be tightly fitted and keyed to their respective shafts. The teeth shall be hardened.

3.11.4 Brakes. Each winch drum and the boom swinger shall be equipped with a brake operated by a foot pedal through linkage. The linkage shall be so arranged that the brakes are released upon release of pressure on the foot pedals. The brake shall give efficient braking without grab, chatter, or overheating under full-load conditions. The brake shall be adjustable with the removal of major components.

3.11.5 Winch controls. Engine, winch and boom swinger controls shall be located on the right hand side of the winch. A saddle seat and steel floor plate platform shall be provided for the operator.

3.11.6 Cables. Unless otherwise specified (see 6.2), cables shall be furnished with the winch and shall conform to RR-W-410, type I, class 2, 6 x 19, construction 3, IWRC, improved plow steel, uncoated, preformed regular right lay. Cable of sufficient length to perform the functions specified in 4.5.2.2 shall be furnished and shall have a breaking strength of not less than four times the force required for such testing.

3.12 Electromagnetic interference. The electromagnetic interference emission characteristcs of the winch shall conform to MIL-STD-461, part 8, class C1, group II, except that a 20 dB relaxation is permitted in the emission limits.

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3.13 Base. The entire winch including the engine shall be mounted on a rigidly constructed, skid type base of structural steel with welded connections. The base shall withstand all specified operational and handling loads of the winch. The top surfaces shall be machined to provide alinement for the side frames and engine.

3.14 Toolbox. The winch shall be equipped with a weatherproof toolbox. The toolbox shall be made of metal not less than 0.071 inch thick and shall accommodate all tools required for maintenance and operation of the winch. The lid shall be secured to one side by a continuous hinge. A hasp shall be fastened to the lid to lock the toolbox. The hasp shall accommodate a padlock having a minimum 3/8-inch shackle. The lid shall open through an arc of at least 90 degrees and shall remain open by its own weight or by holding devices. A drain hole with elastomer plug shall be provided in the bottom of the toolbox.

3.15 Lubricants. Unless otherwise specified herein, the winch shall be serviced with military lubricants selected by the contractor in accordance with table I, SAE J754. Run-in and testing shall be performed with these lubricants as applicable for the ambient-operating temperature. A lubrication data plate identifying military lubricants and applicable temperature ranges shall be provided and mounted (see 3.19.1) on each winch.

3.15.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected accessible position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with handoperated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

3.15.2 Enclosures with integral reservoir of lubricant. Enclosures, such as gearcases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. Drain plugs shall be of the magnetic type. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the winch is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be provided.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.16.1.

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a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.17 Lifting attachments. Permanently affixed attachments that enable the winch to be lifted in its normal travel position shall be provided. Attachments shall conform to MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.18.

3.18 Tiedown attachments. Permanently affixed attachments that permit fastening the winch to the floor or deck of a transportation medium shall be provided. Attachments shall conform to MIL-STD-209. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.17.

3.19 Identification marking. The winch shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.19.1 Instruction plates or labels. Each winch shall be equipped with instruction plates, labels, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the winch. Instruction plates and labels shall conform to MIL-P-514, type III, compositin C of type I, grade A, class 1 material and the labeling requirements of MIL-STD-1472. Plates or labels shall be attached by screws, bolts, or rivets in a location where the plate or label will be both visible and legible.

3.20 Treatment and painting. The portions of the winch and its components and parts normally painted in commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704. When specified (see 6.2), treatment and painting shall be type A. Unless otherwise specified (see 6.2), treatment and painting shall be type F or type G as applicable.

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3.21 Workmanship.

3.21.1 Castings and forgings. All parts, components, and assemblies of the winch which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.21.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly, except in the performance of designed heat treatment. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.21.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the winch to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible, and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. The welding process used in fabrication of the winch shall be at the option of the contractor.

3.21.3.1 Welders. Before assigning any welder to welding work covered by this specification, the contractor shall obtain certification that the welder or automatic welding equipment operator has passed qualification tests as prescribed by one of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Welding Qualifications, Sections 5 and 6.
Welding Qualifications of the ASME, Section IX.
Earthmoving and Construction Equipment Welding, AWS/ANSI D14.3 Qualification, Section 5.

Contractors who only make horizontal welds need not qualify welders for "all position welding." The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

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3.21.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers, or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.21.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492 or IFI Standard No. 114 for Break Mandrel Blind Rivets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and regulations.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article winch shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article winch shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each winch shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each winch shall be tested as specified in 4.5.2.4, 4.5.2.6, and 4.5.2.7. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. Winches shall be examined as specified herein for the following defects:

Defect No.	Defect	Requirement paragraph
101.	Material not as specified.	3.3
102.	Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals not insulated as specified.	3.3.1.1
104.	Bearings not as specified.	3.3.3
105.	Screw threads not as specified.	3.3.5
106.	Safety provisions not as specified.	3.5
107.	Human factors not as specified.	3.6
108.	Noise hazard sign missing or not as specified.	3.6.1
109.	Diesel engine not as specified.	3.9
110.	Alternator not as specified.	3.9.1
111.	Charging receptacle missing or not as specified.	3.9.2
112.	Engine mounting and housing not as specified.	3.9.3
113.	Fuel tank not as specified.	3.9.4
114.	Cranking system not as specified.	3.9.5
115.	Engine controls missing or not as specified.	3.9.6
116.	Instrumentation missing or not as specified.	3.9.7
117.	Engine cold start aid not as specified.	3.9.8
118.	Batteries missing or not as specified.	3.9.9
119.	Battery box not as specified.	3.9.10
120.	Drive clutch, shafting and chain not as specified.	3.10
121.	Winch and boom swinger assembly not as specified.	3.11 thru 3.11.5
122.	Winch factor of safety not as specified.	3.11.1
123.	Cables not as specified.	3.11.6
124.	Winch base not as specified.	3.13
125.	Toolbox missing or not as specified.	3.14
126.	Lubricants not as specified.	3.15
127.	Lubrication data plate missing or not as specified.	3.15
128.	Lubrication fittings missing or not as specified.	3.15.1
129.	Enclosures with integral reservoirs of lubricant not as specified.	3.15.2
130.	Fungus and moisture resistance not as specified.	3.16
131.	Lifting and tiedown attachments missing or not as specified.	3.17, 3.18
132.	Identification marking missing or not as specified.	3.19
133.	Instruction plates missing or not as specified.	3.19.1

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134.	Treatment and painting not as specified.	3.20
135.	Castings and forgings not as specified.	3.21.1
136.	Metal fabrication not as specified.	3.21.2
137.	Welding not as specified.	3.21.3
138.	Bolted connections not as specified.	3.21.4
139.	Riveted connections not as specified.	3.21.5
140.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.1.2
141.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the winch.	3.3.2
142.	Welders certification not as specified.	3.21.3.1
143.	Interchangeability not as specified.	3.3.4

4.5.2 Tests.

4.5.2.1 Engine. The diesel engine shall be tested or certified in accordance with 5.2 and 5.3 of MIL-STD-1410.

4.5.2.2 Performance. The winch shall be anchored. The cables shall be played out for a distance of 75 feet over the boom swinger and attached to a 5,000-pound load. The load shall then be pulled by the winch for a distance of 5 feet and then stopped. The above starting and stopping cycle shall be accomplished not less than 10 times within a 10-minute period. After a 10-minute interval, the above test shall be repeated. The winch shall also exert a single-cable pull of not less than 11,800 pounds at a line speed of 200 feet per minute acting on the second wrap of the cable around the winch drum. Any evidence of malfunction, deformation, or damage to the winch during these tests shall constitute failure of the tests.

4.5.2.3 Environmental.

4.5.2.3.1 Operational temperature. The winch shall be subjected to a temperature of -25° F for a period of not less than 4 hours. After this 4-hour period, the winch shall be operated and used in winching operations at -25° F. This test shall be repeated at a temperature of +115° F. Inability of the winch to operate as specified in 3.4.1 at either extreme temperature shall constitute failure of this test.

4.5.2.3.2 Precipitation. With the engine operating at its continuous rated power, the winch shall be exposed to precipitation for 15 minutes at a rate of 1/4 gallon per minute per square foot of wetted surface impinging on the winch with a 30 mph wind. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.4 Brake. The winch brake shall be tested under maximum load (see 3.11.1) with the winch anchored. The cable shall be attached to the maximum load and the load lifted to a height of 20 feet and then allowed to drop intermittently at regular speed intervals with the brake being applied at each

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interval to stop and hold the load until the brake is released manually. This test shall be repeated 10 times in 15 minutes. Evidence of overheating or inability of the brake to stop and hold the load shall constitute failure of this test.

4.5.2.5 Tiedown and lifting attachments. The static load as specified in 3.17 for lifting attachments and in 3.18 for tiedown attachments shall be applied to the attachment when it is mounted on its structural member. Weld failure or permanent deformation of the attachments or the structural member shall constitute failure of this test.

4.5.2.6 Winch load. The winch line shall be loaded and operated as specified in 3.11.1. The load shall be hoisted vertically to a distance of 20 feet, then lowered slowly at approximately 10 feet per minute. Each time the load descends 1 foot it shall be brought to a complete halt. This cycle of operation shall be repeated 10 times. Nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.7 Boom swinger. The boom swinger shall be loaded and operated as specified in 3.11.2. The boom with the load shall be moved in an arc of approximately 90 degrees. The rate of acceleration and deceleration shall be such that the load line shall not exceed an angle of 10 degrees from the vertical. This test shall be repeated 20 times. Nonconformance to 3.11.2 shall constitute failure of this test.

4.5.2.8 Maintenance operations. The winch shall be examined for accessibility of major components. The accessibility of drains and path of discharge of lubricants and coolants shall be demonstrated. Each operation on the maintenance operations list (see 3.7) shall be performed. Nonconformance to 3.7 and 3.15.2 shall constitute failure of this test.

4.5.2.9 Electromagnetic interference. The winch shall be tested to determine conformance to 3.12. Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.10 Cable breaking strength. Any cable furnished with the winch shall withstand the breaking strength specified in the applicable cable specification. Nonconformance to 3.11.6 shall constitute failure of this test.

4.5.2.11 Noise level measurement. The noise level of the winch shall be measured in accordance with MIL-STD-1474 at the operator's station while the winch is exerting a single-cable pull of not less than 8,000 pounds at a line speed of 200 feet per minute acting on the second wrap of the cable around the winch drum. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

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4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The first article pack of crated winches for level A shall be subjected to the rail impact test in accordance with MIL-STD-1186, appendix A. Subsequent examination and rejection criteria shall be in accordance with the referenced standard.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

144. Disassembly not as specified for level A.
145. Matchmarking not as specified for level A.
146. Surfaces of components requiring a contact preservative in accordance with the referenced document not coated with preservative for level A.
147. Engine and engine accessories not preserved in accordance with the referenced document for level A or C.
148. Exposed gears not preserved as specified for level A.
149. Gear housing not filled and tagged as specified for level A.
150. Drive chains not preserved as specified for level A.
151. Clutches not preserved and secured in the disengaged position as specified for level A.
152. Brakes not preserved as specified for level A.
153. Technical publications not preserved as specified for level A or C.
154. Repair parts not preserved in accordance with referenced document for level A or C.
155. Maintenance tools not preserved in accordance with the referenced document for level A or C.
156. Consolidation not as specified for level A.
157. Packing not as specified for levels A or C.
158. Strapping not zinc coated for level A.
159. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 First article pack. The supplier shall furnish a first article pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to

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surveillance and approval by the Government (see 6.5). The first article pack may be accomplished utilizing either the first article winch or a production winch. If the first article model winch is utilized, any preservation and packing shall be removed by the supplier at no expense to the Government when requested by the Government to facilitate comparison between the first article model winch and the production winch.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss. Bolts, nuts, pins, and washers involved shall be placed on one of the mating parts and secured to prevent their loss.

5.2.1.2 Matchmarking. Parts that are removed shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be on shipping tags conforming to UU-T-81, type A, and shall be applied using a waterproof ink.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces. Unpainted exterior metal surfaces of component parts requiring the application of a preservative in accordance with MIL-P-116, such as bolts, nuts, springs, sheaves, sprockets, pins, linkages, levers, pawl and ratchet level winding devices, lubricating fittings, and the base and any other unprotected metal surfaces not specified herein shall be coated with type P-1 preservative.

5.2.1.5 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062 as specified for type II, method I.

5.2.1.6 Gears.

5.2.1.6.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.6.2 Enclosed gears. Gear housing shall be filled to the operating level with the lubricant recommended for operation and the gears actuated to insure coating all interior parts and surfaces of the housing with the gear lubricant. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating: "The housing is filled to operating level with lubricant required for operation". The information shall be applied using a waterproof ink.

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5.2.1.7 Drive chains.

5.2.1.7.1 Exposed chains. Exposed chains shall be coated with enough type P-3 or P-9 preservative to insure penetration to the inner surfaces of the roller, pins, and bushings. After the excess preservative has drained, the entire chain and the sprockets shall be coated with type P-1 preservative.

5.2.1.7.2 Enclosed chains. Enclosed chains shall be preserved as specified in 5.2.1.6.2 for enclosed gears.

5.2.1.8 Clutches.

5.2.1.8.1 Disc type. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to a minimum. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. The clutch shall be secured in the completely disengaged position.

5.2.1.8.2 Cone-type. The unlined cone and clutch control mechanisms shall be coated with a thin film of primer conforming to TT-P-664. The clutches shall not be operated after application of the primer. Clutch control mechanisms not enclosed with housings shall be coated with type P-1 preservative. The clutch shall be released to prevent the lining from adhering to the unlined cones.

5.2.1.9 Brakes. Facing of the brake drums and brake control linkages shall be coated with a thin film of primer conforming to TT-P-664.

5.2.1.10 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-3.

5.2.1.11 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.12 Maintenance tools. Maintenance tools shall be preserved in accordance with level A requirements of PPP-P-40.

5.2.1.13 Consolidation. Preserved small components shall be placed in the toolbox, as space permits, and the toolbox lid closed and secured. Components that will not fit in the toolbox shall be consolidated together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Blocking, bracing, and anchoring of the components to prevent movement and damage shall be provided. The boxes shall be of a configuration to permit minimum cube and shall be secured to the winch or to the base of the container specified in 5.3, with strapping conforming to QQ-S-781, type I or IV, class B, size as applicable.

5.2.2 Level C.

5.2.2.1 Engine. The engine shall be preserved in accordance with the level C alternate procedure requirement of MIL-E-10062.

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5.2.2.2 Repair parts, maintenance tools, and technical publications. Such repair parts, maintenance tools, and technical publications as are specified shall be preserved as specified for level A and placed in the toolbox. The toolbox lid shall be closed and secured. Any such parts that will not fit in the toolbox shall be consolidated in close fitting weather resistant boxes which shall be secured to the winch with steel strapping.

5.2.2.3 Other components. All other components of the winch shall be preserved to afford protection against deterioration and damage from the supplier to the initial destination. The supplier's standard practice is acceptable when it fulfills these requirements.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each complete winch, preserved and consolidated as specified in 5.2, shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style a. Blocking, bracing, anchoring, closure, and strapping shall be in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186. Strapping shall be zinc-coated.

5.3.2 Level C. Each winch, preserved as specified in 5.2, shall be packed in a manner which will prevent deterioration, damage, and pilferage during shipment and short term storage. As a minimum the winch shall be protected with a shroud of suitable material which shall be secured to the winch so as to prevent loosening, flapping, tearing or loss.

5.4 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The winch is intended for use as an all-purpose, portable type hoist for general use with a 30-ton stiff-leg derrick.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is not required (see 3.2).
c. When first article is not required (see 3.2).
d. When engine cold starting aid will not be furnished (see 3.9.8).
e. When winch head line guide is not to be furnished (see 3.11.1.5).
f. When cables are not required or will be other than as specified (see 3.11.6).
g. When type F or type G treatment and painting is required (see 3.20).
h. Time frame required for submission of the first article pack (see 5.1).
i. Level of preservation and packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in

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procurement documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first article(s).

6.4 Noise. When the noise level of the winch is 85dB(A) or greater, the appropriate discussion of the noise hazard per MIL-STD-1474 (MANUALS Paragraph) shall be incorporated in the manuals and should include the requirement for hearing protection, the noise level of the winch, the distance from the winch at which 85dB(A) will always be met, and what operational situations or system configuration will reduce the level to 85dB(A).

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the supplier of his obligation to preserve, pack, and mark the winches in accordance with this specification.

Custodians:
Army - ME
Navy - YD

Preparing activity:
Army - ME

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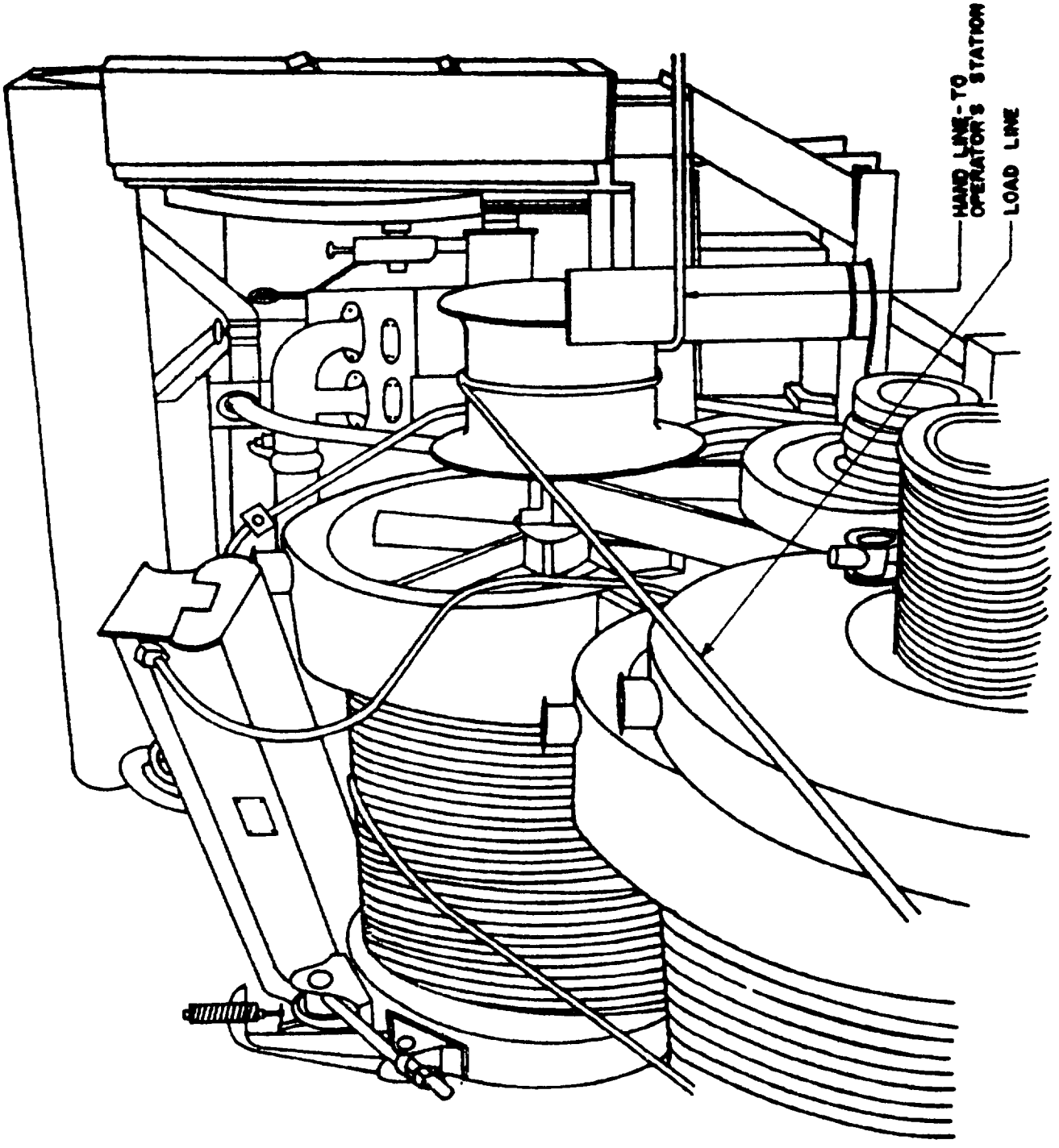


FIGURE 1 - WINCH HEAD ...E GUIDE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-694C	Winch, Diesel-Driven, 2-Drum, 5.9 Ton Pull, with Boom Swinger

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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